Investigating and Comparing Discussion Topics in Multilingual Underground Forums

List of IT-words used to enrich for cluster filtering

To filter out clusters that mainly contain *non-conversational text*, we rely on an English dictionary that was enriched with the set of well-known IT words shown in Table 1.

Table 1: Well-known IT words we enriched our dictionary with.

Well known IT words
admin, apikey, array, attacks, austria, autosizes, bytes, captcha, change, chat, chris, client, columheaders, czechia, dateline, dos, ddos, debug, description, displayorder, dns, dport, drop, email, encrypt, event, facebook, false, filter, following, forumid, forward, frame, freebsd, function, germany, get, gmail, gnulinux, hash, height, hpux, icmp, icq, iframe, input, ipaddress, iptables, irix, jabber, kernel, key, keyarg, keycode, keylogger, keys, let, linux, listening, loopback, martin, masquerade, md5, microsoft, mysql, nat, netherlands, options, output, pass, pass, php, pluto, policy, port, postid, postrouting, powershell, prerouting, pro, product, public, quick, random, redirect, registration,return, romania, root, rsa, script, server, skype, smb, smp, sms, src, startup, state, states, stratus, sudo, switzerland, taglist, tcp, threadid, timeout, title, trendmicro, type, udp, unicode, united, url, usergroupid, userid, username, var, varname, visible, vmware, windows, x64, xml, zynamics

Common Topics

Table 2 shows information about the cluster pairs that have *highly related* topics. Each row pictures a Russian language cluster (left side) and a corresponding highly related English language cluster (right side). A cluster can have more that one highly related cluster from the other data set.

Table 2: Discovered *common topics* (cluster pairs with *high similarity*). P=Paragraphs, T=Threads, A=Authors. Note[b]: Some threads (< 1%) have a missing author.

Russian language dataset				**English language dataset**			
Number of				**Number of**			
P	**T**	**A**[b]	**Cluster label**	**P**	**T**	**A**[b]	**Cluster label**
880	737	546	'buy_mail_base'	293	241	177	'buy_mailpass_mail'
1428	968	730	'need_telegram_bot'	426	327	261	'telegram_telegram_bot'
272	270	260	'looking_seller_sellers_buyer'	147	147	122	'pm_pm_buyers'
370	347	291	'buy_database_databases'	515	409	254	'selling_database_data'
70	64	40	'iframe_form_payment'	33	32	19	'cc_form_payment_payments'
3398	2594	1751	'buy_base_bases'	223	175	153	'buy_base_bases'
630	418	349	'redirect_redirects'	168	120	102	'redirect_redirects_link'

5122	2709	1709	’domains_domains_domain’	576	359	294	’domains_domains_domain’
880	737	546	’buy_mail_base’	137	119	100	’buy_emails_email_business’
6765	3435	2340	’need_crypt_encryption_file’	1558	878	700	’crypt_crypter_encryption’
305	289	260	’price_price_account’	222	199	173	’pay_price_account_balance’
1068	780	508	’buy_database_site_sites’	147	123	101	’dating_base_site_email’
2500	1907	1278	’need_language_translation’	185	168	153	’speak_russian_language_people’
829	740	583	’buy_traffic_countries’	436	316	260	’buy_traffic_quality’
454	379	332	’need_mail_hack’	89	73	62	’need_email_emails’
51	48	48	’buy_base_email_sale’	137	119	100	’buy_emails_email_business’
306	212	190	’need_file_pdf’	829	516	382	’exploit_exploit_office_word’
172	134	103	’buy_loader_play’	105	73	59	’play_loader’
267	206	173	’sms_spam’	300	230	173	’sms_spam’
182	151	134	’sms_sms_interception’	82	62	59	’sms_sms_interception’
287	215	192	’need_calls_number’	253	161	144	’need_calls_service_voice’
61	55	49	’grabber_grabber_form’	26	24	23	’formgrabber_requests_internet’
1521	1033	748	’buy_browser_chrome’	468	289	232	’firefox_browser_internet_explorer’
997	740	577	’buy_paypal_account’	374	268	221	’paypal_paypal_account’
233	211	183	’buy_vbv_cvv’	81	68	58	’buy_cvv_shop’
271	267	228	’work_guarantor_forum’	515	487	333	’work_guarantor_forum’
1997	1949	1499	’write_contacts_pm’	165	159	138	’contact_contacts’
532	483	429	’looking_coder_team’	382	311	255	’looking_coder_developer’
727	382	305	’looking_miner_mining’	221	113	95	’looking_miner_mining’
836	760	665	’test_test_tests’	228	203	182	’test_test_tests’
54	43	41	’selling_license_smoke_bot’	75	59	51	’sell_license_smoke_loader’
1004	498	391	’need_tor_network’	153	98	89	’tor_tor_network’
2809	1409	1017	’need_proxy_proxies’	488	315	265	’proxy_proxies’
1555	1101	832	’buy_ads_adwords’	760	492	344	’looking_ads_adwords’
3205	2122	1400	’buy_exploit_vulnerability’	830	567	425	’exploit_exploit_exploits’
71	66	60	’address_email_phone_format’	1812	1218	731	’need_scans_email_phone’
35	35	31	’address_number_phone_birth’	1812	1218	731	’need_scans_email_phone’
90	74	70	’buy_email_mail’	87	65	58	’checker_emails_email’
267	206	173	’sms_spam’	567	381	303	’sms_sms_bulk’
400	329	289	’sms_sms_number’	567	381	303	’sms_sms_bulk’
1521	1033	748	’buy_browser_chrome’	26	24	23	’formgrabber_requests_internet’
103	92	74	’access_access_domain_admin’	279	237	149	’access_access_admin_domain’
2983	1680	1206	’buy_cards_card’	661	461	380	’need_cards_card_credit’
122	117	106	’buy_yusa_sale’	293	241	177	’buy_mailpass_mail’
167	159	149	’buy_mix_world’	436	316	260	’buy_traffic_quality’
196	196	175	’price_guarantor’	515	487	333	’work_guarantor_forum’
27	27	19	’form_form_site’	100	96	52	’shop_orders_day’
287	260	225	’buy_downloads_countries’	436	316	260	’buy_traffic_quality’
254	210	169	’buy_accounts_amazon’	269	191	143	’looking_accounts_amazon_aws’
297	184	124	’card_card_video’	89	56	47	’geforce_geforce_card_mb’
54	48	28	’redirect_payment’	55	48	16	’redirect_paypal_payments_payment’
1665	1124	785	’works_windows_system’	156	145	118	’works_windows_x64’
498	429	395	’return_money_moneyback_refund’	135	114	102	’refund_refund_money’
231	230	211	’send_pm_contact’	280	278	221	’pm_jabber’
1323	1248	999	’apologize_topic_section’	37	36	36	’sorry_thread_section_place’
1822	1526	1146	’write_jabber_pm’	165	159	138	’contact_contacts’
83	74	63	’buy_mustache_pm’	147	113	91	’buy_loads_price’
83	74	63	’buy_mustache_pm’	1599	1151	736	’logs_logs_log’
67	50	41	’looking_networks_network’	46	32	31	’looking_pentester_pentesters’

157	157	6	’rights_pcs_ip_user’	59	59	6	’access_pc_ip_scanner’
361	249	218	’keylogger_keylogger_keyboard’	224	135	117	’keylogger_keylogger_family’
143	105	86	’buy_forms_tax’	206	148	115	’buy_forms_tax’
518	320	278	’need_machine_vmware’	117	90	84	’vmware_workstation_guest’
1321	960	791	’looking_botnet_botnets’	447	324	268	’looking_botnet_botnets’
38	27	23	’selling_access_office_version’	170	135	94	’office_office_office365’
163	132	127	’need_invite_invites’	88	57	50	’need_invite_genesis’
167	151	129	’support_proxy_socks5’	521	341	285	’socks_socks_socks5’
232	183	139	’buy_gifts_gift’	133	86	75	’card_card_gift’
33	30	28	’looking_traffic_adwords’	760	492	344	’looking_ads_adwords’
859	452	350	’dns_dns_server’	119	95	85	’server_dns_domain’
270	222	198	’buy_vps_server’	260	209	180	’need_vps_bulletproof’
164	162	141	’strait_date_year’	144	78	70	’date_6th_time’
701	554	486	’sale_server_servers’	65	57	50	’rent_rent_month’
829	740	583	’buy_traffic_countries’	147	113	91	’buy_loads_price’
52	50	36	’buy_mile_hash’	293	241	177	’buy_mailpass_mail’
880	737	546	’buy_mail_base’	275	194	148	’leads_leads_email’
880	737	546	’buy_mail_base’	114	75	56	’looking_mailpass_mail’
880	737	546	’buy_mail_base’	223	175	153	’buy_base_bases’
880	737	546	’buy_mail_base’	1599	1151	736	’logs_logs_log’
51	48	48	’buy_base_email_sale’	293	241	177	’buy_mailpass_mail’
63	56	53	’works_office_versions’	98	59	46	’office_office_bit’
423	292	240	’looking_macro_office’	829	516	382	’exploit_exploit_office_word’
578	472	408	’need_newsletter_email’	64	57	52	’need_emails_inbox_day’
136	120	107	’looking_coder_android’	32	29	26	’looking_developer_android_coder’
136	120	107	’looking_coder_android’	382	311	255	’looking_coder_developer’
292	229	202	’buy_number_phone_numbers’	44	36	27	’need_database_phone_number’
371	287	234	’need_python_coder’	269	121	110	’looking_python_programming’
557	523	464	’support_support_service’	63	56	54	’support_support_customer’
937	848	736	’offer_service_services’	53	52	48	’pm_service_services’
216	159	139	’uac_uac_bypass’	105	85	77	’bypass_uac’
2260	1816	1294	’need_panel_admin’	279	237	149	’access_access_admin_domain’
65	48	44	’sell_keys_windows’	32	22	20	’windows_key_product’
155	149	131	’time_days_order’	59	53	50	’delivery_hours_time’
31	29	25	’buy_database_mail’	137	119	100	’buy_emails_email_business’
91	79	68	’buy_dumps_dump’	223	175	153	’buy_base_bases’
2108	1845	1427	’countries_countries_country’	242	215	195	’countries_countries_country’
165	162	147	’write_forum_pm’	165	159	138	’contact_contacts’
77	76	70	’payment_transactions_guarantor_forum’	515	487	333	’work_guarantor_forum’
133	132	121	’work_guarantor_pm’	515	487	333	’work_guarantor_forum’
502	434	391	’discounts_discount’	133	110	104	’discount_discount_discounts_orders’
229	203	189	’buy_installs_installes’	160	123	102	’buy_installs_countries’
30	30	30	’know_topic_flame’	37	36	36	’sorry_thread_section_place’
138	110	92	’buy_accounts_seller’	269	191	143	’looking_accounts_amazon_aws’
51	46	45	’buy_chase_mail’	211	161	122	’logs_logs_bank’
774	646	513	’buy_banks_bank’	211	161	122	’logs_logs_bank’
198	141	97	’processor_processor_ghz’	226	111	90	’cpu_cores_processor_ghz’
933	631	516	’accept_money_qiwi_yandex’	30	23	23	’money_wallet_qiwi_yandex’
54	48	28	’redirect_payment’	33	32	19	’cc_form_payment_payments’
44	44	43	’accept_webmoney_payment’	418	375	325	’accept_btc_payment’
595	456	408	’selling_license_licenses’	172	126	112	’license_license_updates’

231	230	211	’send_pm_contact’	233	233	162	’contact_telegram_pm’
35	34	33	’ask_people_answers_hint’	270	257	211	’help_help_advance’
1822	1526	1146	’write_jabber_pm’	280	278	221	’pm_jabber’
40	39	37	’looking_partners_partner’	915	783	594	’looking_money_term_partner’
1857	1632	1265	’looking_software_program’	69	64	63	’software_software_program’
83	74	63	’buy_mustache_pm’	223	175	153	’buy_base_bases’
2101	1540	1091	’buy_ftp_server’	274	200	170	’ftp_ftp_server’
102	102	40	’end_problem_guarantor_auction’	515	487	333	’work_guarantor_forum’
76	75	54	’end_hours_bet’	25	25	19	’end_hours_bet’
51	48	35	’blitz_blitz_bet_hours’	25	25	19	’end_hours_bet’
138	111	100	’looking_terminal_pos_terminals’	80	64	54	’pos_access_sale’
2500	1907	1278	’need_language_translation’	76	65	59	’language_language_languages’
1729	957	701	’buy_money_poker_casino’	130	101	84	’looking_database_casino_poker’
62	59	53	’price_lines_line’	110	95	70	’lines_lines_line’
464	306	247	’flash_flash_player’	156	89	72	’player_exploit_flash’
136	116	111	’crack_crack_program’	33	25	25	’looking_software_internet’
92	88	73	’write_gills_pm_gill’	165	159	138	’contact_contacts’
146	127	118	’buy_access_network_networks’	189	161	122	’selling_access_admin_network’
192	152	91	’buy_access_citrix’	114	106	55	’citrix_access’
742	551	435	’buy_vnc_bot’	447	324	268	’looking_botnet_botnets’
742	551	435	’buy_vnc_bot’	184	111	96	’looking_hvnc_software’
317	226	184	’buy_hvnc_bot’	64	49	40	’android_rat’
317	226	184	’buy_hvnc_bot’	447	324	268	’looking_botnet_botnets’
62	61	59	’offer_slag_pm’	147	147	122	’pm_pm_buyers’
228	201	188	’subscription_subscription_ subscriptions’	65	57	50	’rent_rent_month’
69	58	51	’buy_cpanel_access’	169	126	96	’buy_cpanels_cpanel’
69	58	51	’buy_cpanel_access’	293	241	177	’buy_mailpass_mail’
83	70	67	’looking_botnet_android’	148	111	87	’looking_bot_android’
623	507	396	’looking_bot_android’	64	49	40	’android_rat’
623	507	396	’looking_bot_android’	148	111	87	’looking_bot_android’
623	507	396	’looking_bot_android’	447	324	268	’looking_botnet_botnets’
87	73	67	’list_bots_ip’	119	95	85	’server_dns_domain’
3443	2158	1569	’looking_bot_bots’	733	431	341	’looking_bot_bots’
2722	2092	1388	’buy_soap_soaps’	293	241	177	’buy_mailpass_mail’
365	315	279	’looking_stream_streams’	65	57	50	’rent_rent_month’
306	248	207	’buy_traffic_iframe’	436	316	260	’buy_traffic_quality’
308	262	222	’need_access_computer_control’	104	90	83	’rdp_rdp_server_access’
35	24	21	’need_fax_faxes’	567	381	303	’sms_sms_bulk’
51	51	48	’send_link_pm_links’	165	159	138	’contact_contacts’
53	48	40	’sell_root_server’	189	161	122	’selling_access_admin_network’
403	362	325	’buy_dedics_sale’	223	175	153	’buy_base_bases’
477	390	268	’buy_response_job’	218	171	135	’dump_dumps’
477	390	268	’buy_response_job’	436	316	260	’buy_traffic_quality’
477	390	268	’buy_response_job’	275	194	148	’leads_leads_email’
477	390	268	’buy_response_job’	223	175	153	’buy_base_bases’
538	332	236	’buy_channels_youtube’	107	67	55	’buy_channels_youtube’
3398	2594	1751	’buy_base_bases’	275	194	148	’leads_leads_email’
160	128	106	’alexa_rank_traffic’	134	88	72	’rank_rank_country’
264	224	184	’buy_traffic_android’	160	123	102	’buy_installs_countries’
264	224	184	’buy_traffic_android’	29	27	25	’android_traffic’
207	197	180	’throw_people_time’	181	162	138	’know_method_methods’

2809	1409	1017	’need_proxy_proxies’	521	341	285	’socks_socks_socks5’
310	167	137	’selling_tunnels_ssh_tunnel’	208	135	113	’ssh_ssh_tunnel’
270	190	137	’yandex_site’	168	120	102	’redirect_redirects_link’
149	123	87	’buy_logs_facebook’	275	194	148	’leads_leads_email’
149	123	87	’buy_logs_facebook’	1599	1151	736	’logs_logs_log’
113	97	85	’buy_accounts_facebook_business’	275	194	148	’leads_leads_email’
66	62	52	’visitors_visitors_day’	45	41	36	’visitors_visits_day’
43	36	32	’need_email_flood’	36	25	19	’flooder_service_email_flood’
27	23	20	’need_hotmail_mail’	89	73	62	’need_email_emails’
51	50	44	’ability_files_file_mail’	70	57	50	’mailer_script_attachment_email’
880	737	546	’buy_mail_base’	218	171	135	’dump_dumps’
880	737	546	’buy_mail_base’	436	316	260	’buy_traffic_quality’
880	737	546	’buy_mail_base’	170	135	94	’office_office_office365’
51	48	48	’buy_base_email_sale’	147	123	101	’dating_base_site_email’
51	48	48	’buy_base_email_sale’	96	63	53	’bins_bins_bin’
63	56	53	’works_office_versions’	156	145	118	’works_windows_x64’
1659	1297	1029	’looking_spam_spammer’	997	715	520	’need_spammer_spam’
211	169	160	’looking_exe_pdf’	829	516	382	’exploit_exploit_office_word’
104	97	94	’need_letters_inbox_day’	64	57	52	’need_emails_inbox_day’
286	162	98	’need_phone_nokia’	153	126	99	’iphone_windows_gb’
46	42	36	’need_sms_bot’	82	62	59	’sms_sms_interception’
121	77	71	’viber_viber_numbers’	69	46	43	’whatsapp_whatsapp_way_message’
476	392	304	’buy_android_pm’	160	123	102	’buy_installs_countries’
476	392	304	’buy_android_pm’	436	316	260	’buy_traffic_quality’
292	229	202	’buy_number_phone_numbers’	219	163	126	’need_numbers_phone’
349	272	230	’need_sms_numbers’	567	381	303	’sms_sms_bulk’
200	171	162	’need_site_web’	157	132	116	’looking_website_web_developer’
64	51	46	’looking_loader_loaders’	229	181	172	’loader_loader_file’
1521	1033	748	’buy_browser_chrome’	83	68	63	’firefox_passwords_password_ browsers’
227	200	155	’buy_shells_traffic_day’	436	316	260	’buy_traffic_quality’
787	653	492	’buy_shells_shell’	436	316	260	’buy_traffic_quality’
787	653	492	’buy_shells_shell’	203	158	126	’buy_shells_shell’
241	200	176	’screenshots_screenshots_screen’	31	27	26	’screenshot_screenshot_screenshots_ desktop’
124	124	118	’pass_test_guarantor’	515	487	333	’work_guarantor_forum’
467	407	293	’selling_access_admin_panel’	189	161	122	’selling_access_admin_network’
887	540	413	’need_cookies_cookie’	166	132	118	’need_cookies_cookie’
186	151	126	’passwords_passwords_browsers’	83	68	63	’firefox_passwords_password_ browsers’
89	82	63	’find_password_admin_administrator’	279	237	149	’access_access_admin_domain’
482	369	309	’need_brute_force’	230	184	144	’need_force_checker’
44	32	31	’bypass_smartscreen_windows_filter’	36	29	23	’bypass_smartscreen’
357	270	227	’need_scan_scanner’	189	146	127	’scanner_scanner_scanning’
131	85	79	’allows_signature_signatures’	317	165	141	’ssl_certificate_signing’
370	347	291	’buy_database_databases’	218	171	135	’dump_dumps’
370	347	291	’buy_database_databases’	293	241	177	’buy_mailpass_mail’
91	79	68	’buy_dumps_dump’	515	409	254	’selling_database_data’
91	79	68	’buy_dumps_dump’	293	241	177	’buy_mailpass_mail’
91	79	68	’buy_dumps_dump’	1599	1151	736	’logs_logs_log’
45	36	29	’looking_language_programmer_ developer’	29	19	13	’need_coder_project’

34	29	23	’need_stock_banks_bank’	36	26	14	’need_bank_account’
152	129	107	’need_rights_root_server’	111	79	57	’escalation_command_privilege’
122	117	106	’buy_yusa_sale’	137	119	100	’buy_emails_email_business’
122	117	106	’buy_yusa_sale’	223	175	153	’buy_base_bases’
167	159	149	’buy_mix_world’	147	113	91	’buy_loads_price’
165	162	147	’write_forum_pm’	87	87	81	’contact_pm_information’
648	498	396	’installation_installation_installer’	72	67	64	’installs_installer_installation’
271	267	228	’work_guarantor_forum’	505	457	310	’work_escrow_payment’
271	267	228	’work_guarantor_forum’	24	24	19	’work_guarantor_forum’
313	310	285	’work_prepayment_guarantor’	515	487	333	’work_guarantor_forum’
92	90	83	’transaction_transactions_guarantor’	515	487	333	’work_guarantor_forum’
270	266	237	’sell_sale_hand_hands’	44	44	37	’sell_hand_hands’
272	270	260	’looking_seller_sellers_buyer’	147	113	91	’buy_loads_price’
329	244	176	’create_file_configuration’	55	32	31	’intecom_checklistxlsx_configuration_file’
51	44	29	’redirect_orders_day’	48	46	29	’redirect_redirect_orders_shop’
654	517	270	’orders_orders_day’	100	96	52	’shop_orders_day’
229	203	189	’buy_installs_installes’	436	316	260	’buy_traffic_quality’
98	90	85	’buy_mix_world’	126	97	92	’buy_installs_install’
98	90	85	’buy_mix_world’	436	316	260	’buy_traffic_quality’
287	260	225	’buy_downloads_countries’	147	113	91	’buy_loads_price’
287	260	225	’buy_downloads_countries’	1599	1151	736	’logs_logs_log’
134	126	122	’buy_installations_installs_exchange’	126	97	92	’buy_installs_install’
254	210	169	’buy_accounts_amazon’	211	161	122	’logs_logs_bank’
543	492	415	’buy_accounts_account’	147	147	122	’pm_pm_buyers’
543	492	415	’buy_accounts_account’	253	208	178	’buying_accounts_account’
51	46	45	’buy_chase_mail’	170	135	94	’office_office_office365’
51	46	45	’buy_chase_mail’	293	241	177	’buy_mailpass_mail’
162	161	141	’need_advice_tips’	270	257	211	’help_help_advance’
1008	723	588	’loader_loader_panel’	229	181	172	’loader_loader_file’
93	85	76	’startup_entry_registry_program’	79	62	56	’startup_persistence’
123	114	98	’form_payment_site’	33	32	19	’cc_form_payment_payments’
363	319	290	’withdraw_money_withdrawal_account’	43	29	26	’withdraw_withdrawal_funds’
31	31	22	’form_form_payment_site’	33	32	19	’cc_form_payment_payments’
42	41	38	’accept_wmz_payment’	418	375	325	’accept_btc_payment’
498	429	395	’return_money_moneyback_refund’	596	394	303	’paid_money_days_guy’
349	316	287	’scammers_time_scam’	505	457	310	’work_escrow_payment’
176	162	155	’wrote_pm_time’	73	62	57	’stopped_messages_days’
152	147	137	’contact_questions_pm’	87	87	81	’contact_pm_information’
1770	1452	1133	’said_money_person_time’	596	394	303	’paid_money_days_guy’
231	230	211	’send_pm_contact’	76	76	69	’contact_pm’
231	230	211	’send_pm_contact’	111	107	85	’contact_pm’
196	155	132	’agreed_defendant_work’	596	394	303	’paid_money_days_guy’
76	76	70	’looking_information_info’	270	257	211	’help_help_advance’
72	69	63	’help_assistance_advance’	270	257	211	’help_help_advance’
81	73	69	’looking_programmer_c_coder’	382	311	255	’looking_coder_developer’
1822	1526	1146	’write_jabber_pm’	233	233	162	’contact_telegram_pm’
48	48	10	’guarantor_people_expense’	515	487	333	’work_guarantor_forum’
536	450	377	’need_ligament_ligaments’	148	111	87	’looking_bot_android’
61	50	33	’buy_luxsocks_account’	45	30	26	’buy_luxsocks_sale’
83	74	63	’buy_mustache_pm’	436	316	260	’buy_traffic_quality’

83	74	63	’buy_mustache_pm’	515	409	254	’selling_database_data’
83	74	63	’buy_mustache_pm’	275	194	148	’leads_leads_email’
83	74	63	’buy_mustache_pm’	293	241	177	’buy_mailpass_mail’
83	74	63	’buy_mustache_pm’	38	30	26	’buy_logs_botnet’
83	74	63	’buy_mustache_pm’	253	208	178	’buying_accounts_account’
424	293	242	’need_captcha_site_service’	53	46	44	’captcha_captcha_login_page’
2101	1540	1091	’buy_ftp_server’	189	161	122	’selling_access_admin_network’
179	136	117	’looking_strike_cobalt’	175	114	90	’strike_strike_cobalt’
49	49	43	’end_drop_time_auction’	32	32	20	’availability_pm_auction_hours’
51	48	35	’blitz_blitz_bet_hours’	90	90	53	’end_bid_auction’
200	196	132	’end_auction_hours’	59	52	34	’end_hours_bid’
200	196	132	’end_auction_hours’	90	90	53	’end_bid_auction’
44	44	27	’end_hours_auction’	55	55	28	’end_hours_bid’
160	140	124	’looking_stealer_azorult’	447	324	268	’looking_botnet_botnets’
1068	780	508	’buy_database_site_sites’	218	171	135	’dump_dumps’
1068	780	508	’buy_database_site_sites’	275	194	148	’leads_leads_email’
1068	780	508	’buy_database_site_sites’	137	119	100	’buy_emails_email_business’
1068	780	508	’buy_database_site_sites’	293	241	177	’buy_mailpass_mail’
55	50	46	’buy_sheets_sheet’	223	175	153	’buy_base_bases’
62	59	53	’price_lines_line’	222	199	173	’pay_price_account_balance’
1477	1036	807	’need_drops_drop’	374	268	221	’paypal_paypal_account’
52	30	29	’written_burp_suite’	40	33	30	’suite_columns_burp’
684	503	409	’buy_grandfathers_grandfather’	223	175	153	’buy_base_bases’
1009	852	696	’rent_bunch_month’	65	57	50	’rent_rent_month’
586	550	481	’unsubscribe_unsubscribe_pm_topic’	181	162	138	’know_method_methods’
146	127	118	’buy_access_network_networks’	114	106	55	’citrix_access’
146	127	118	’buy_access_network_networks’	170	135	94	’office_office_office365’
146	127	118	’buy_access_network_networks’	293	241	177	’buy_mailpass_mail’
162	116	101	’buy_teamviewer_team’	51	44	42	’teamviewer_viewer_team’
162	116	101	’buy_teamviewer_team’	121	96	87	’vnc_vnc_session’
1768	1139	767	’buy_rdp_access’	208	135	113	’ssh_ssh_tunnel’
1768	1139	767	’buy_rdp_access’	104	90	83	’rdp_rdp_server_access’
742	551	435	’buy_vnc_bot’	80	64	54	’pos_access_sale’
317	226	184	’buy_hvnc_bot’	184	111	96	’looking_hvnc_software’
294	250	218	’looking_pentesters_team’	46	32	31	’looking_pentester_pentesters’
172	148	132	’need_site_sites’	422	302	232	’need_page_phishing’
136	90	80	’need_ip_cloudflare’	45	32	31	’bypass_cloudflare_ip’
149	124	107	’buy_logs_cookies’	31	28	25	’logs_logs_botnet’
149	124	107	’buy_logs_cookies’	170	135	94	’office_office_office365’
149	124	107	’buy_logs_cookies’	211	161	122	’logs_logs_bank’
62	61	59	’offer_slag_pm’	223	175	153	’buy_base_bases’
87	72	59	’selling_logs_zeus’	38	30	26	’buy_logs_botnet’
28	25	22	’sell_license_andromeda_time’	75	59	51	’sell_license_smoke_loader’
69	58	51	’buy_cpanel_access’	269	191	143	’looking_accounts_amazon_aws’
69	58	51	’buy_cpanel_access’	189	161	122	’selling_access_admin_network’
888	507	384	’need_ddos_attacks’	131	94	86	’ddos_attack_service’
83	70	67	’looking_botnet_android’	447	324	268	’looking_botnet_botnets’
623	507	396	’looking_bot_android’	149	106	87	’looking_botnet_android’
140	120	108	’looking_bot_bank’	136	104	83	’looking_bot_banking’
1321	960	791	’looking_botnet_botnets’	148	111	87	’looking_bot_android’
83	65	57	’looking_report_credit’	155	59	48	’report_report_credit_score’
2722	2092	1388	’buy_soap_soaps’	170	135	94	’office_office_office365’

306	248	207	’buy_traffic_iframe’	1599	1151	736	’logs_logs_log’
308	262	222	’need_access_computer_control’	121	96	87	’vnc_vnc_session’
204	114	97	’ransomware_ransomware_malware’	276	164	137	’ransomware_ransomware_ransom’
26	23	19	’set_password_cookies_login’	170	135	94	’office_office_office365’
221	139	100	’looking_powershell_script’	93	66	59	’powershell_command’
35	24	21	’need_fax_faxes’	300	230	173	’sms_spam’
51	51	48	’send_link_pm_links’	233	233	162	’contact_telegram_pm’
51	51	48	’send_link_pm_links’	280	278	221	’pm_jabber’
38	37	36	’build_50_price’	24	23	21	’cost_build_builder’
38	37	36	’build_50_price’	222	199	173	’pay_price_account_balance’
403	362	325	’buy_dedics_sale’	293	241	177	’buy_mailpass_mail’
403	362	325	’buy_dedics_sale’	1599	1151	736	’logs_logs_log’
158	130	113	’need_apk_pm’	72	52	45	’crypt_apk_crypter’
477	390	268	’buy_response_job’	160	123	102	’buy_installs_countries’
477	390	268	’buy_response_job’	66	56	49	’buy_rdp_quality’
477	390	268	’buy_response_job’	203	158	126	’buy_shells_shell’
477	390	268	’buy_response_job’	137	119	100	’buy_emails_email_business’
477	390	268	’buy_response_job’	40	27	17	’buy_info_profiles’
477	390	268	’buy_response_job’	147	113	91	’buy_loads_price’
477	390	268	’buy_response_job’	1599	1151	736	’logs_logs_log’
1126	845	704	’program_program_affiliate’	915	783	594	’looking_money_term_partner’
3398	2594	1751	’buy_base_bases’	293	241	177	’buy_mailpass_mail’
272	221	206	’buy_links_link’	1599	1151	736	’logs_logs_log’
264	224	184	’buy_traffic_android’	436	316	260	’buy_traffic_quality’
25	23	20	’takes_minutes_installation_server’	53	44	41	’uptime_server_hours_time’
131	119	109	’buy_sox_server’	521	341	285	’socks_socks_socks5’
1004	498	391	’need_tor_network’	42	29	24	’looking_traffic_onion_onions’
114	39	36	’fastflux_fastflux_networks’	119	95	85	’server_dns_domain’
114	39	36	’fastflux_fastflux_networks’	66	33	28	’fastflux_fastflux_panel’
612	355	277	’buy_ssh_server’	208	135	113	’ssh_ssh_tunnel’
2417	940	706	’vpn_vpn_server’	153	98	89	’tor_tor_network’
178	174	153	’looking_suppliers_supplier’	38	37	36	’looking_vendor_suppliers_supplier’
232	183	139	’buy_gifts_gift’	96	63	53	’bins_bins_bin’
96	69	47	’buy_itunes_gift’	133	86	75	’card_card_gift’
421	357	302	’buy_socks_socks5’	66	56	49	’buy_rdp_quality’
421	357	302	’buy_socks_socks5’	521	341	285	’socks_socks_socks5’
421	357	302	’buy_socks_socks5’	223	175	153	’buy_base_bases’
130	121	116	’bypass_bypass_bypasses’	69	64	63	’software_software_program’
90	88	71	’selling_access_sale’	189	161	122	’selling_access_admin_network’
149	123	87	’buy_logs_facebook’	436	316	260	’buy_traffic_quality’
149	123	87	’buy_logs_facebook’	137	119	100	’buy_emails_email_business’
61	30	24	’allows_telnet_client’	103	84	77	’smtp_smtp_server’
145	74	55	’interception_traffic_network’	140	107	88	’network_packets_nmap’
320	229	188	’need_sniffer_sniff’	62	44	43	’sniffer_sniffer_cc’
58	55	53	’depends_price_traffic’	215	209	182	’price_price_quality’
270	222	198	’buy_vps_server’	576	359	294	’domains_domains_domain’
75	71	70	’need_hosting_bulletproof’	260	209	180	’need_vps_bulletproof’
154	131	121	’servers_servers_countries’	242	215	195	’countries_countries_country’
829	740	583	’buy_traffic_countries’	126	97	92	’buy_installs_install’
829	740	583	’buy_traffic_countries’	160	123	102	’buy_installs_countries’
829	740	583	’buy_traffic_countries’	66	56	49	’buy_rdp_quality’
829	740	583	’buy_traffic_countries’	203	158	126	’buy_shells_shell’

829	740	583	'buy_traffic_countries'	275	194	148	'leads_leads_email'
829	740	583	'buy_traffic_countries'	1599	1151	736	'logs_logs_log'
227	182	142	'looking_ssn_number'	1812	1218	731	'need_scans_email_phone'
28	26	25	'sell_base_ssn'	293	241	177	'buy_mailpass_mail'
28	26	25	'sell_base_ssn'	1812	1218	731	'need_scans_email_phone'
77	62	53	'buy_accounts_developer_play'	34	25	21	'accounts_account_developer_apple'
52	50	36	'buy_mile_hash'	137	119	100	'buy_emails_email_business'
40	38	15	'date_date_release'	144	78	70	'date_6th_time'
42	37	33	'offer_services_classmates_networks'	89	73	62	'need_email_emails'
36	34	33	'depends_price_mail'	64	57	52	'need_emails_inbox_day'
100	80	73	'need_spam_spammer'	27	22	19	'looking_users_discord_server'
100	80	73	'need_spam_spammer'	426	327	261	'telegram_telegram_bot'
49	46	44	'agree_product_terms_rules'	105	88	83	'agree_right_service_rules'
32	31	30	'need_script_mail'	40	29	24	'need_shell_sendmail_mail'
27	23	20	'need_hotmail_mail'	25	22	20	'need_inbox_gmail'
87	78	76	'help_password_mail'	27	27	21	'access_access_email'
149	141	132	'need_access_mail'	27	27	21	'access_access_email'
156	137	123	'pass_database_mail_mailpass'	114	75	56	'looking_mailpass_mail'
156	137	123	'pass_database_mail_mailpass'	77	68	63	'need_list_email_mail'
156	137	123	'pass_database_mail_mailpass'	293	241	177	'buy_mailpass_mail'
107	96	90	'substitution_email_sender'	64	57	52	'need_emails_inbox_day'
880	737	546	'buy_mail_base'	147	123	101	'dating_base_site_email'
880	737	546	'buy_mail_base'	269	191	143	'looking_accounts_amazon_aws'
880	737	546	'buy_mail_base'	66	56	49	'buy_rdp_quality'
880	737	546	'buy_mail_base'	219	163	126	'need_numbers_phone'
880	737	546	'buy_mail_base'	83	79	43	'mailpass_usa_mail'
880	737	546	'buy_mail_base'	147	113	91	'buy_loads_price'
880	737	546	'buy_mail_base'	71	64	49	'buy_fullz_sale'
51	48	48	'buy_base_email_sale'	71	64	49	'buy_fullz_sale'
1659	1297	1029	'looking_spam_spammer'	138	100	71	'looking_office_spammer'
1659	1297	1029	'looking_spam_spammer'	64	57	52	'need_emails_inbox_day'
180	172	154	'buy_base_spam_bases'	293	241	177	'buy_mailpass_mail'
99	83	76	'looking_addresses_email_mail'	87	65	58	'checker_emails_email'
950	673	510	'buy_smtp_server'	87	65	58	'checker_emails_email'
950	673	510	'buy_smtp_server'	64	57	52	'need_emails_inbox_day'
950	673	510	'buy_smtp_server'	247	192	150	'looking_smtp_inbox'
105	78	52	'buy_logs_gmail'	293	241	177	'buy_mailpass_mail'
105	78	52	'buy_logs_gmail'	1599	1151	736	'logs_logs_log'
161	154	145	'reserve_right_explanation_service'	105	88	83	'agree_right_service_rules'
198	153	120	'buy_skrill_neteller'	374	268	221	'paypal_paypal_account'
210	186	163	'buy_links_logs'	1599	1151	736	'logs_logs_log'
121	77	71	'viber_viber_numbers'	426	327	261	'telegram_telegram_bot'
336	249	197	'need_apk_play'	69	56	55	'need_apk_play'
476	392	304	'buy_android_pm'	29	27	25	'android_traffic'
476	392	304	'buy_android_pm'	1599	1151	736	'logs_logs_log'
54	53	51	'looking_logs_people'	915	783	594	'looking_money_term_partner'
52	52	50	'price_price_log'	215	209	182	'price_price_quality'
182	151	134	'sms_sms_interception'	567	381	303	'sms_sms_bulk'
400	329	289	'sms_sms_number'	300	230	173	'sms_spam'
120	112	102	'sold_copies_copy'	24	23	21	'cost_build_builder'
61	55	49	'grabber_grabber_form'	468	289	232	'firefox_browser_internet_explorer'
214	188	168	'looking_services_web_development'	32	29	26	'looking_developer_android_coder'

214	188	168	'looking_services_web_development'	382	311	255	'looking_coder_developer'
214	188	168	'looking_services_web_development'	157	132	116	'looking_website_web_developer'
27	24	23	'buy_lock_browser_browsers'	184	111	96	'looking_hvnc_software'
260	226	194	'looking_code_javascript'	31	24	21	'need_coder_javascript'
227	200	155	'buy_shells_traffic_day'	203	158	126	'buy_shells_shell'
227	200	155	'buy_shells_traffic_day'	147	113	91	'buy_loads_price'
787	653	492	'buy_shells_shell'	275	194	148	'leads_leads_email'
787	653	492	'buy_shells_shell'	189	161	122	'selling_access_admin_network'
787	653	492	'buy_shells_shell'	293	241	177	'buy_mailpass_mail'
787	653	492	'buy_shells_shell'	147	113	91	'buy_loads_price'
787	653	492	'buy_shells_shell'	223	175	153	'buy_base_bases'
787	653	492	'buy_shells_shell'	1599	1151	736	'logs_logs_log'
290	170	114	'need_sound_mp3'	229	148	91	'audio_formats_video'
937	848	736	'offer_service_services'	447	324	268	'looking_botnet_botnets'
937	848	736	'offer_service_services'	171	153	129	'looking_hacker_hackers'
937	848	736	'offer_service_services'	68	63	61	'tools_tools_tool'
379	346	186	'company_services_industry'	258	232	138	'company_access_industry'
997	740	577	'buy_paypal_account'	253	208	178	'buying_accounts_account'
73	68	58	'users_users_database'	515	409	254	'selling_database_data'
449	366	316	'looking_locker_lockers'	148	111	87	'looking_bot_android'
1575	1054	776	'need_password_passwords'	632	456	379	'need_password_login'
106	68	59	'buy_certificate_code_signing'	317	165	141	'ssl_certificate_signing'
151	142	137	'terms_time_days_hours'	59	53	50	'delivery_hours_time'
31	29	25	'buy_database_mail'	293	241	177	'buy_mailpass_mail'
31	29	25	'buy_database_mail'	223	175	153	'buy_base_bases'
370	347	291	'buy_database_databases'	147	123	101	'dating_base_site_email'
370	347	291	'buy_database_databases'	189	161	122	'selling_access_admin_network'
370	347	291	'buy_database_databases'	223	175	153	'buy_base_bases'
370	347	291	'buy_database_databases'	44	44	37	'sell_hand_hands'
91	79	68	'buy_dumps_dump'	218	171	135	'dump_dumps'
91	79	68	'buy_dumps_dump'	269	191	143	'looking_accounts_amazon_aws'
91	79	68	'buy_dumps_dump'	147	147	122	'pm_pm_buyers'
45	36	29	'looking_language_programmer_developer'	382	311	255	'looking_coder_developer'
327	190	130	'injection_injection_sql'	290	159	116	'injection_injection_sql'
156	118	98	'set_rights_file'	27	18	17	'chrootmysqldevnull_permissions_file_tmd'
99	84	74	'buy_bins_base'	96	63	53	'bins_bins_bin'
233	211	183	'buy_vbv_cvv'	218	171	135	'dump_dumps'
233	211	183	'buy_vbv_cvv'	137	119	100	'buy_emails_email_business'
233	211	183	'buy_vbv_cvv'	293	241	177	'buy_mailpass_mail'
122	117	106	'buy_yusa_sale'	218	171	135	'dump_dumps'
122	117	106	'buy_yusa_sale'	275	194	148	'leads_leads_email'
122	117	106	'buy_yusa_sale'	1599	1151	736	'logs_logs_log'
62	61	56	'buy_loading_cis'	436	316	260	'buy_traffic_quality'
167	159	149	'buy_mix_world'	275	194	148	'leads_leads_email'
167	159	149	'buy_mix_world'	154	124	108	'mix_loads_world'
2108	1845	1427	'countries_countries_country'	154	124	108	'mix_loads_world'
1428	968	730	'need_telegram_bot'	27	22	19	'looking_users_discord_server'
1428	968	730	'need_telegram_bot'	69	46	43	'whatsapp_whatsapp_way_message'
90	88	84	'share_pm_people'	69	68	59	'contact_pm_telegram_thanks'
33	15	15	'ram_mbits_month'	194	68	65	'ram_ram_traffic'

165	162	147	’write_forum_pm’	176	171	144	’pm_price’
165	162	147	’write_forum_pm’	76	76	69	’contact_pm’
165	162	147	’write_forum_pm’	233	233	162	’contact_telegram_pm’
165	162	147	’write_forum_pm’	280	278	221	’pm_jabber’
42	42	40	’work_guarantor_exploit’	515	487	333	’work_guarantor_forum’
196	196	175	’price_guarantor’	24	24	19	’work_guarantor_forum’
192	191	179	’work_service_guarantor_expense’	24	24	19	’work_guarantor_forum’
192	191	179	’work_service_guarantor_expense’	515	487	333	’work_guarantor_forum’
133	132	121	’work_guarantor_pm’	24	24	19	’work_guarantor_forum’
48	48	48	’discounts_discounts_wholesalers’	133	110	104	’discount_discount_discounts_orders’
272	270	260	’looking_seller_sellers_buyer’	269	191	143	’looking_accounts_amazon_aws’
272	270	260	’looking_seller_sellers_buyer’	275	194	148	’leads_leads_email’
272	270	260	’looking_seller_sellers_buyer’	247	192	150	’looking_smtp_inbox’
272	270	260	’looking_seller_sellers_buyer’	80	74	67	’price_code_source’
272	270	260	’looking_seller_sellers_buyer’	253	208	178	’buying_accounts_account’
272	270	260	’looking_seller_sellers_buyer’	82	72	63	’buy_banks_bank’
139	136	127	’price_price_piece’	126	97	92	’buy_installs_install’
606	595	524	’price_price_prices’	215	209	182	’price_price_quality’
654	517	270	’orders_orders_day’	24	24	12	’orders_orders_site’
85	76	57	’orders_orders_day_card’	100	96	52	’shop_orders_day’
85	76	57	’orders_orders_day_card’	24	24	12	’orders_orders_site’
273	239	203	’download_file’	76	65	58	’download_file’
229	203	189	’buy_installs_installes’	275	194	148	’leads_leads_email’
98	90	85	’buy_mix_world’	160	123	102	’buy_installs_countries’
287	260	225	’buy_downloads_countries’	203	158	126	’buy_shells_shell’
287	260	225	’buy_downloads_countries’	275	194	148	’leads_leads_email’
43	41	37	’looking_person_installations_partner’	915	783	594	’looking_money_term_partner’
134	126	122	’buy_installations_installs_exchange’	436	316	260	’buy_traffic_quality’
134	126	122	’buy_installations_installs_exchange’	147	147	122	’pm_pm_buyers’
111	106	97	’pay_installs_price’	126	97	92	’buy_installs_install’
111	106	97	’pay_installs_price’	160	123	102	’buy_installs_countries’
111	106	97	’pay_installs_price’	222	199	173	’pay_price_account_balance’
111	106	97	’pay_installs_price’	215	209	182	’price_price_quality’
396	364	308	’buy_accounts_logs’	253	208	178	’buying_accounts_account’
68	68	65	’buy_acc_pm’	147	147	122	’pm_pm_buyers’
254	210	169	’buy_accounts_amazon’	170	135	94	’office_office_office365’
337	331	285	’buy_akki_aki’	269	191	143	’looking_accounts_amazon_aws’
51	46	45	’buy_chase_mail’	269	191	143	’looking_accounts_amazon_aws’
51	46	45	’buy_chase_mail’	253	208	178	’buying_accounts_account’
39	19	19	’gbits_anlym_mb_ram’	194	68	65	’ram_ram_traffic’
39	19	19	’gbits_anlym_mb_ram’	226	111	90	’cpu_cores_processor_ghz’
2115	1951	1312	’write_toad_pm’	176	171	144	’pm_price’
2115	1951	1312	’write_toad_pm’	165	159	138	’contact_contacts’
241	238	216	’write_suggestions_pm’	57	54	52	’appreciated_advice_suggestions’
51	50	41	’accept_payment_liberty’	418	375	325	’accept_btc_payment’
123	114	98	’form_payment_site’	418	375	325	’accept_btc_payment’
123	114	98	’form_payment_site’	55	48	16	’redirect_paypal_payments_payment’
70	64	40	’iframe_form_payment’	55	48	16	’redirect_paypal_payments_payment’
119	110	103	’system_payments_payment_systems’	418	375	325	’accept_btc_payment’
1665	1124	785	’works_windows_system’	98	59	46	’office_office_bit’
43	36	33	’compressed_file_compression’	170	135	100	’bytes_kb_size_file’
123	103	90	’unpack_archive_zip’	202	65	52	’need_file_zip’

150	145	136	’write_experience_people’	57	54	52	’appreciated_advice_suggestions’
152	147	137	’contact_questions_pm’	76	76	69	’contact_pm’
231	230	211	’send_pm_contact’	40	40	36	’pm_pm_person’
231	230	211	’send_pm_contact’	165	159	138	’contact_contacts’
231	230	211	’send_pm_contact’	87	87	81	’contact_pm_information’
134	131	119	’salary_salary_month’	65	57	50	’rent_rent_month’
1997	1949	1499	’write_contacts_pm’	87	87	81	’contact_pm_information’
1997	1949	1499	’write_contacts_pm’	76	76	69	’contact_pm’
1997	1949	1499	’write_contacts_pm’	233	233	162	’contact_telegram_pm’
1997	1949	1499	’write_contacts_pm’	280	278	221	’pm_jabber’
81	73	69	’looking_programmer_c_coder’	70	52	41	’need_coder_php’
532	483	429	’looking_coder_team’	32	29	26	’looking_developer_android_coder’
532	483	429	’looking_coder_team’	29	19	13	’need_coder_project’
79	64	58	’looking_code_programmer_ programming’	382	311	255	’looking_coder_developer’
199	166	151	’training_courses_course’	228	203	182	’test_test_tests’
69	69	62	’looking_person_cooperation’	915	783	594	’looking_money_term_partner’
1822	1526	1146	’write_jabber_pm’	87	87	81	’contact_pm_information’
1822	1526	1146	’write_jabber_pm’	76	76	69	’contact_pm’
1822	1526	1146	’write_jabber_pm’	117	108	98	’support_account_jabber_server’
480	193	132	’detained_information_years_code’	120	53	46	’said_attacks_government_security’